A New Metric Based on Association Rules to Assess Feature-Attribution Explainability Techniques for Time Series Forecasting

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***Abstract*—This paper introduces a new, model-independent, metric, called RExQUAL, for quantifying the quality of explanations provided by attribution-based explainable artificial intelligence techniques and compare them. The underlying idea is based on feature attribution, using a subset of the ranking of the attributes highlighted by a model-agnostic explainable method in a forecasting task. Then, association rules are generated using these key attributes as input data. Novel metrics, including global support and confidence, are proposed to assess the joint quality of generated rules. Finally, the quality of the explanations is calculated based on a wise and comprehensive combination of the association rules global metrics. The proposed method integrates local explanations through attribution-based approaches for evaluation and feature selection with global explanations for the entire dataset. This paper rigorously evaluates the new metric by comparing three explainability techniques: the widely used SHAP and LIME, and the novel methodology RULEx. The experimental design includes predicting time series of different natures, including univariate and multivariate, through deep learning models. The results underscore the efficacy and versatility of the proposed methodology as a quantitative framework for evaluating and comparing explainable techniques.**

***Index Terms*—Explainable artificial intelligence, attribution-based explainability, association rules, time series forecasting.**

I. Introduction

THE field of Artificial Intelligence (AI) has become a powerful tool that transforms the technological landscape in various fields such as healthcare, finance, transportation, and time series analysis. As AI advances, its influence is likely to intensify, and its impact continues to shape industries, improving efficiency, innovation, and the way people interact with technology [3]. However, despite the remarkable capabilities of Machine Learning (ML) and Deep Learning (DL) models, their inherent complexity leads to a lack of transparency and interpretability. DL models, in particular, often function as intricate black boxes, obscuring the rationale behind their decisions and posing challenges for gaining trust and acceptance from researchers and domain experts. This opacity is especially problematic in critical applications such as healthcare, financial predictions, or personal data processing, where understanding the reasoning behind decisions is crucial. Consequently, as reliance on AI grows, there is an increasing demand for transparent and interpretable models. This field, known as eXplainable Artificial Intelligence (XAI), aims to provide a range of techniques that enable human users to understand, trust, and create more explainable models, enhancing transparency and offering insights into decision-making processes. XAI, along with related concepts such as trustworthy AI, understandable AI, and interpretability, advocates for the development of models that improve human understanding through clear reasoning [45]. In the landscape of XAI, various typologies distinguish between global and local interpretability, model-dependent versus model-agnostic approaches, and different degrees of granularity in explanation techniques. First, global interpretability refers to understanding a model's overall behavior and providing insights into its general decision-making processes. In contrast, local interpretability focuses on specific predictions, offering a more detailed understanding of the model's rationale for individual instances. Secondly, XAI methods can be categorized based on their adaptability to specific models or ability to explain predictions across various techniques. Model-agnostic approaches are versatile, applicable to various models, and promote flexibility. These methods, known as post-hoc interpretability, are applied after training to explain predictions [4]. One of the most widely used XAI techniques is feature attribution, which refers to determining the contribution or importance of individual input features in influencing the output of an ML model [26]. Although incorporating XAI techniques represents a significant step toward interpretability, a crucial question emerges. Researchers wonder about the effectiveness of explanations provided by the XAI systems when presented to human users. The central focus is on evaluating the quality of explanations, user satisfaction, comprehension of AI systems, and overall system performance. Therefore, the challenge lies in developing explainable models

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and establishing robust methodologies to evaluate and compare the effectiveness of various XAI techniques [14].

Thus, this paper introduces Rule-based Explanation QUALity (RExQUAL), a novel metric designed for the quantitative evaluation of XAI techniques. Local explanations are utilized through attribution-based methods for feature evaluation and selection, while global explanations are derived by applying the selected features to provide insights into the overall behavior of the model. In this way, there is a lack of standardized metrics specifically designed to evaluate the quality of these XAI techniques. To the best of our knowledge, no existing metric comprehensively measures the effectiveness of XAI methods in providing meaningful and accurate explanations. RExQUAL aims to fill this gap by offering a quantitative approach to assess the quality of explanations. It is a general-purpose metric to assess the quality of explanations obtained by attribution-based explainability methods and could be applied to explain predictions obtained from any ML model and any type of data. However, the experimental framework for the evaluation of this methodology has been focused on the prediction of time series since almost all scientific disciplines consist of data sampled over time and the topic of time series forecasting has been receiving increasing attention in recent years, especially because of its interdisciplinary nature and the large amount of temporal data that is nowadays generated [12], [38]. The main contributions of this work are described below.

- Introducing RExQUAL, an objective methodology for quantitatively evaluating attribution-based XAI techniques. This methodology provides a robust framework for systematically assessing the performance and interpretability of various XAI methods.
- Integrating local and global XAI, using attribution-based feature selection to derive insights at both instance and dataset levels.
- Presenting RExQUAL as an open source tool, available in a GitHub repository. The methodology is accessible to the broader research community, enabling researchers to apply, modify, and extend the tool for evaluating the quality of explanations generated by XAI techniques.[1]
- Defining new global quality metrics for sets of association rules, addressing a critical gap in current evaluation methods.
- Positioning RExQUAL as a tool for standard comparison across attribution-based XAI techniques, allowing researchers to objectively measure and contrast the quality of explanations produced by different XAI methods.
- Conducting a comparative evaluation of attribution-based XAI technologies, including SHAP [20], LIME [46], and RULEx [43], specifically in the context of time series forecasting.
- Applying the methodology to real-world time series data from diverse domains, such as energy management and agriculture. By testing the explainability techniques in these critical areas, we demonstrate the practical utility of the proposed approach.

The remainder of the paper is structured as follows. First, current research in evaluating XAI is presented in Section II. The methodology of the novel approach presented in this paper is then detailed in Section III. After that, the methodology is tested with different examples using time series data in Section IV. Finally, the results are discussed in Section V, and the conclusions are presented in Section VI.

II. Related Work

XAI is vital in fields like spatial epidemiology [41], medicine [2], [24], [27], [47], remote sensing [16], [40], agriculture [25], [32], environment [18], and energy [5], enabling better model transparency and fostering trust in decision-making. Evaluating XAI methods is essential for understanding their effectiveness, yet researchers have identified a lack of consensus and standardization in the classification and assessment of these techniques. The literature predominantly emphasizes two perspectives: human-centered evaluation, often termed subjective analysis, and computer-centered objective evaluation. However, the subjective nature and varying quality of some analyses present significant challenges [19]. Furthermore, in relation to the ethical dimensions of AI and its social implications, several reviews have proposed metrics to provide insights into the outputs of various AI models, as discussed in the comprehensive study by [37].

A. Subjective Evaluation Metrics

The first approach to evaluating XAI draws inspiration from cognitive states in psychology. For example, the authors in [11] presented a comparative analysis of two subjective XAI methodologies which are representations of users' internal understanding of the system. Similarly, the researchers in [15] proposed a set of seven subjective cognitive metrics to evaluate XAI systems. In addition, the article by [33] introduced an objective measurement idea combined with human assessment, proposing four metrics: performance differences (D) between the explanation's logic and the agent's actual performance, the number of rules (R) generated, the number of features (F) used, and the stability (S) of the explanation. These metrics were presented theoretically. Similarly, [29] introduced various approaches to evaluate XAI methods, outlining the requirements for an effective evaluation system. The proposed metrics assess different technical aspects of the methods, such as "faithfulness," which evaluates how well an explanation aligns with true feature contributions (measured by Pearson correlation), "monotonicity," which ensures consistency in the relative importance of features, and "incompleteness," which measures the explanation's resilience to noise and perturbations. Furthermore, in the study conducted by [34], the authors explored XAI methods that typically rely on visual interpretability when applied to time series data. The results demonstrated the robust performance of SHAP. Additionally, researchers in [36] addressed the challenge of XAI in healthcare. They introduced definitions for two XAI measurements: similarity and stability, to clarify the concept of

[1]The source code can be accessed at the following link: https://github.com/Data-Science-Big-Data-Research-Lab/rexqual.

TABLE I
TABLE SUMMARIZING XAI EVALUATION METHODS FROM SECTION II, DETAILING THE EVALUATION OF EXPLANATIONS BASES

Reference	Basis	Nature	Approach	Experiments	Kind of data	Evaluation
[11]	Human-centered	Subj	Theoretical	No	Not specified	Comparison between XAI techniques
[15]	Human-centered	Subj	Theoretical	No	Not specified	Human evaluation
[33]	Hybrid	Obj	Theoretical	No	Not specified	Comparison, stability, feature contributions
[29]	Hybrid	Obj	Theoretical	No	Not specified	Stability, feature contributions
[34]	Hybrid	Obj	Theoretical and experimental	Yes	Medical images	Expert knowledge
[36]	Hybrid	Obj	Theoretical and experimental	Yes	Time series in healthcare	Accuracy, expert knowledge
[8]	Computer-based	Obj	Experimental	Yes	Image	Attribution
[7]	Computer-based	Obj	Experimental	Yes	Image	Attribution
[30]	Computer-based	Obj	Experimental	Yes	Image	Attribution
[1]	Computer-based	Obj	Theoretical	No	Tabular	Framework
[10]	Computer-based	Obj	Theoretical	No	Tabular	Similarity between explanations
[21]	Computer-based	Obj	Theoretical	No	Tabular, classification tasks	Global scale, post hoc

explainability specifically for time series medical data. Similarity is characterized as a form of accuracy in which explanations provided by the XAI method are compared against annotations created by domain experts. The stability measure evaluates the consistency of explanations through the Pearson correlation coefficient, specifically assessing the correlation between features from the explanations generated by an XAI method and those from alternative XAI techniques. The experiments focus on electrocardiogram (ECG) signals, represented as images.

B. Image Attribution Methods Evaluation

Alternatively, several works in the literature address the issue of evaluating the explanations provided by image classification attribution methods. The paper in [8] addresses the problem of inaccurate regions and poor attribution for misclassified samples. The authors performed a subset selection task, using a function and four constraints (confidence, effectiveness, consistency, and collaboration) to improve attribution quality. Researchers in [7] similarly proposed the Visual Precision Search method, which segments inputs into sub-regions and utilizes consistency and collaboration scores to enhance decision-making localization. Moreover, the authors of [30] introduced RISE, an Explainable AI method for DL models processing images. It generated an importance map by probing DL models with masked images.

C. Evaluation Frameworks

In addition, some frameworks have been created in this domain. The OpenXAI tool, introduced in [1], is an open-source tool designed to standardize the evaluation of post-hoc explanation methods. It implements 22 quantitative metrics that assess the faithfulness, stability, and fairness of XAI methods. However, its initial release focused solely on tabular datasets. A similar tool, Xplique [10], is a library that can be integrated with leading ML libraries such as TensorFlow and incorporates a variety of established XAI methods, though it does not provide the flexibility to implement custom XAI methods. In addition, the researchers in [21] discussed the evaluation of XAI techniques for classification methods. They introduced a novel metric, the Mean Evaluation of Metrics Change (MEMC), which is designed to evaluate the performance of XAI techniques on a global scale.

D. Evaluation Methods Comparison

Finally, Table I summarizes the XAI evaluation techniques that have been introduced in this section. The XAI evaluation methods were compared based on various criteria, including their foundational basis (human or computer-based), nature (objective or subjective), approach (theoretical or experimental), experimental validation, types of data utilized, and specific evaluation metrics employed by the authors. Some methods, like those proposed by [11] and [15], were human-centered with a theoretical approach, focusing on comparing XAI techniques without experimental validation. In contrast, others, such as [34] and [36], adopted a hybrid basis, with evaluations applied to real-world data, particularly in healthcare and medical imaging. Meanwhile, computer-based methods, like [1] and [10], relied solely on objective theoretical frameworks, focusing on tabular data. Finally, there are some interesting approaches such as [7], [8], [30] centered on image attribution explanations. The literature review highlights a noticeable gap in the lack of experimentally demonstrated objective metrics for time series data, whereas such metrics have been extensively developed and validated for image data. The work presented in this paper aims to address this gap by developing an objective and experimentally validated metric for evaluating XAI techniques applied to time series data.

III. METHODOLOGY

This article presents a novel methodology designed for the quantitative and objective evaluation of attribution-based explainability techniques. Furthermore, the proposed approach is characterized by its model independence, indicating its applicability to a wide range of XAI techniques. REXQUAL is a novel metric designed to evaluate the quality of explanations generated by common attribution methods, including LIME, SHAP, and RULEX. While these methods provide local feature attributions for individual predictions, REXQUAL does not perform the attribution itself as it serves as an evaluation tool to assess the effectiveness and reliability of the explanations produced by these methods. The goal of REXQUAL is to ensure that the generated explanations accurately reflect the model's decision-making process. This distinction between attribution and evaluation is critical, as REXQUAL is not directly attributing features to the data. The essential phases of the methodology are visually depicted in Fig. 1 and are detailed further in this section. Specifically, the preliminary steps align with the standard procedure for experimentation in ML tasks: initially, the input data is preprocessed and, in the case of a time series $X_t = \{x_1, \ldots, x_t\}$, a windowing step is required, resulting in an input dataset $X^{N,W+H}$, with N instances, W features and H classes, where the features are the W past values

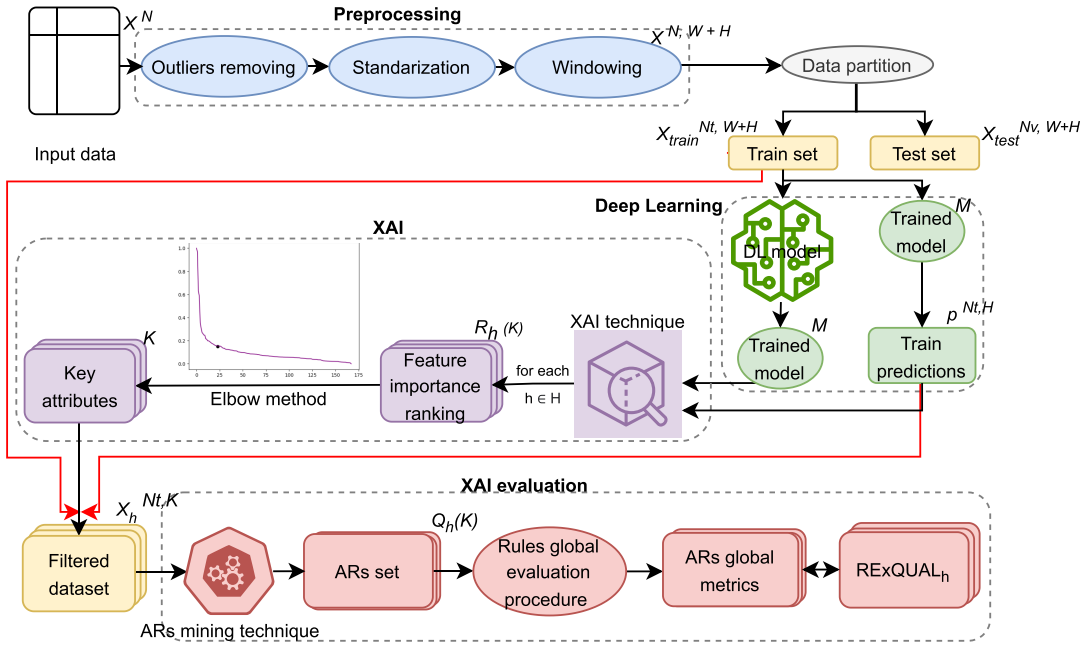


Fig. 1. Flowchart of the methodology. The process is described in Section III: Preprocessing (III-A), DL (III-B), XAI (III-C) and XAI evaluation (III-D).

of the time series and H refers to the prediction horizon, that is, the H next values to be predicted. Subsequently, the data is partitioned into the training set $X_{train}^{N_t,W+H}$ and the test set $X_{test}^{N_v,W+H}$, with N_t and N_v instances, respectively, and a DL model, M, is obtained using the training set. Afterward, the model M generates predictions of the H next values for each instance of the training set, $P^{N_t,H}$, to illustrate the model's output generation process. Next, a XAI technique is employed to extract explanations for the model's decisions. In the case of time series data, explanations are provided for each value h of the prediction horizon H. Therefore, the output of the XAI technique is a ranking of the W features according to their contribution to the expected output for each horizon, denoted as $R_h(W)$. From this ranking, a selection of the K key features with $K \leq W$ is made. Following this, a refined input dataset $X_h^{N_t,K}$ is generated, incorporating only the K most crucial features identified by the XAI technique. Subsequently, a specific Association Rules (ARs) mining technique extracts a set of ARs for each horizon, denoted as $Q_h(K)$, from this refined dataset. Finally, following the global ARs evaluation procedure, proposed global metrics for association rule sets are computed, and the proposed RExQUAL evaluation metric is then calculated for each horizon based on these assessments. A numerical example demonstrating this calculation process is provided in Appendix B, available online. This example illustrates the step-by-step application of the methodology.

A. Preprocessing

The proposed RExQUAL metric is inherently model-independent, allowing for its application across various XAI techniques. However, it is crucial to acknowledge that while the methodology is agnostic to the specific model employed, its effectiveness may exhibit variability depending on the underlying data characteristics. Therefore, RExQUAL is designed to be adaptable and generic, making it suitable for a diverse range of applications. The initial phase involves standard dataset preparation procedures, including outlier identification and management, missing value detection or imputation, and normalization. The preprocessing stage depends on the nature and characteristics of the input data, making it difficult to define a precise approach in advance. The specifics of this process may vary, and in some cases, it can be omitted if the data have already undergone prior preprocessing. Each dataset requires tailored techniques to ensure that the preprocessing aligns with its unique attributes and objectives.

Concerning time series data, a specific number of past events W, also called a time window, are used to predict a particular number of future steps H, the prediction horizon. For this purpose, a windowing process is necessary. Consequently, the time series data are transformed into tabular data, $X^{N,W+H}$, where W represents the size of the windows, and H denotes the prediction horizon. That is,

$$X^{N,W+H} = \{(x^1, y^1), \ldots, (x^N, y^N)\} \; x^i \in \mathbb{R}^W \; y^i \in \mathbb{R}^H \tag{1}$$

where:

$$x^i = \{x_{j+1}, \ldots, x_{j+W}\} \tag{2}$$

$$y^i = \{x_{j+W+1}, \ldots, x_{j+W+H}\} \tag{3}$$

with $x_j \in X_t$. These W attributes are then used to predict the subsequent H values of the time series. Following this preliminary stage, the input data for the DL model are ready to be used. This marks the transition from the initial data processing steps to the subsequent model training and evaluation phase.

B. Deep Learning

DL models are utilized across different problem domains, with the choice of model heavily dependent on the nature of the input data and the specific characteristics of the problem. This phase of the methodology allows for the consideration and implementation of any DL or ML model, providing flexibility in model selection based on the problem at hand. The input data is partitioned using a standard partitioning scheme, typically allocating 70% of the dataset for training and 30% for testing. This strategy is generally effective for large datasets and contributes to the robustness and generalization of the obtained results [13]. However, the partitioning method may need customization for specific application domains, particularly in scenarios with limited data. The training and testing sets are defined as follows:

$$X_{train}^{N_t,W+H} = \left\{(x^1, y^1), \ldots, \left(x^{N_t}, y^{N_t}\right)\right\} \tag{4}$$

$$X_{test}^{N_v,W+H} = \left\{(x^{N_t+1}, y^{N_t+1}), \ldots, \left(x^{N_t+N_v}, y^{N_t+N_v}\right)\right\} \tag{5}$$

having a size of N_t and N_v instances, respectively. The model is trained using the training set data, and the performance of the trained model is assessed with the test set. It is essential to note that, in the context of this paper, the emphasis is not on obtaining accurate predictions but rather on understanding the model's behavior and aspects of interpretability. Consequently, the trained model M is used to generate the predictions $P^{N_t,H}$ of the values $\{y^1, \ldots, y^{N_t}\}$, using $X_{train}^{N_t,W+H}$ as input data. The process of obtaining predictions employs the same data used to train the model to understand the model's inner behavior. This fact is a way to reveal the reasons behind the model's training reasoning and provides insights into the decision-making process.

C. XAI

Next, a post-hoc agnostic XAI technique is applied to explain the model's predictions, meaning that different XAI techniques can be employed. In this aspect, the presented methodology employs a model-independent XAI approach to ensure flexibility across various DL models, making the choice of a post-hoc XAI technique deliberate as it is applied after the prediction process.

Typically, XAI techniques receive input from the trained model M and the data intended for explanation, where these data correspond to the train set $X_{train}^{N_t,W+H}$ in this specific scenario, and the predictions obtained by the model $P^{N_t,H}$. Utilizing the training set as the input for explanation underscores the intention to delve into the factors influencing the predictions. Lastly, the result of the XAI technique is the list of the W features with the importance value of each feature. That means that the XAI technique computes the level of significance of each feature in influencing the prediction of the model $P^{N_t,H}$, for each value $h \in H$. In the case of time series data, the outcome is presented as a ranking $R_h(W)$ for each $h \in H$. Otherwise, there is only one ranking $R(W)$ for non-time series data.

$$R_h\left(\{x_{j+1}, \ldots, x_{j+W}\}\right) = < x_{j+1}^*, \ldots, x_{j+W}^* > \tag{6}$$

where $r(x_{j+1}^*) \geq \ldots \geq r(x_{j+W}^*)$, being r a function of relevance. The calculation of the relevance values varies depending on the specific XAI technique used.

Based on the ranking $R_h(W)$, the most important features are at the top of the ranking. These key features are a subset of $R_h(W)$, considered essential for the predictive performance of the DL model in the forecasting task. Let $R_h(K)$ be the K first features according to its relevance in $R_h(W)$. That is:

$$R_h(K) = < x_{j+1}^*, \ldots, x_{j+K}^* > \tag{7}$$

Therefore, a crucial aspect of the methodology involves carefully selecting the number of these key characteristics K from the ranking with all the W features. This decision plays a significant role in the subsequent phase of the process, considering that the identified top features will be used in the next stage.

The first approach was opting for a predefined number K of top attributes (K = 3, 5, 10...), which may lack appropriateness when dealing with input datasets that exhibit substantial variations in the number of features W. In certain cases, the specified number could prove insufficient or excessive based on the number of characteristics of the input dataset.

Subsequently, the focus shifts to the importance of the values themselves as a means to select the K key features. The approach involves setting a threshold, above which the features that exceed a certain importance value are recognized as key features. First, the importance values undergo normalization to facilitate a more robust comparison between different XAI techniques, and the features are organized according to these normalized values. The next critical step is to select the importance value threshold, which, in this case, is determined using the elbow method. Typically used in cluster analysis, the elbow method helps identify the optimal number of clusters in a dataset. The primary goal is pinpointing where adding more clusters no longer significantly improves the explained variance. This point is referred to as the "elbow" and signifies a balance between model fit and simplicity. In this context, the elbow method establishes the threshold for identifying the desired importance values.

D. XAI Evaluation

The concluding segment of the methodology fulfills the primary objective of this paper: quantitatively measuring the quality of the explanations provided by XAI techniques while also enabling a comparative analysis among them.

In this phase, the essential features $R_h(K)$, recognized in the previous step, are used to refine the training set $X_{train}^{N_t,W+H}$, which originally constituted the input data for the DL model. The training set encompasses N_t samples, the initial number of features W and H values to be predicted. The training set undergoes filtration, retaining only the K key features selected in $R_h(K)$. Additionally, the predictions $P^{N_t,H}$ generated by the DL model are individually incorporated as a class, leading to a newly derived dataset, $X_h^{N_t,K+1}$, for each value h of the prediction horizon H. The $X_h^{N_t,K+1}$ dataset retains the same number of instances but only includes the K key attributes and

the predictions for a single horizon h.

$$X_h^{N_t,K} = \{(x^1, y^1), \ldots, (x^{N_t}, y^{N_t})\}\ x^i \in \mathbb{R}^K\ y^i \in \mathbb{R} \quad (8)$$

where:

$$x^i = \{x^*_{j+1}, \ldots, x^*_{j+K}\} \quad (9)$$

$$y^i = \{x_{j+W+h}\} \quad (10)$$

with $x_j \in X_t$ and $h \in \{1, \ldots, H\}$.

Ultimately, each of these datasets $X_h^{N_t,K+1}$ serves as input for a specific ARs mining technique. In this work, the well-known Apriori algorithm was employed to extract association rules; however, other ARs mining techniques could have been utilized. All sets of rules must be extracted using the same ARs mining technique configured with uniform parameters, ensuring a consistent basis for evaluating the outputs of the XAI techniques for the given input data.

Moreover, the output of the ARs mining algorithm comprises a set of ARs, denoted as $Q_h(K)$ for each value h of the prediction horizon, where the K key features serve as the antecedent and the predictions for the value h of the prediction horizon is forced to be in the consequent. evaluation of an attribution-based XAI technique using association rules (ARs) involves analyzing the relationship between the generated rules and the input data attributes. ARs capture patterns and associations among variables, serving as a means to elucidate how the model's outputs relate to specific input attributes. By delineating dependencies and interactions among these attributes, the ARs reveal nuanced connections between the input characteristics K and the predicted variable. Furthermore, including a variable in the rule antecedent underscores its importance in influencing the outcome, thereby shedding light on the intricate relationships within the data.

Based on this concept, the use of the refined dataset $X_h^{N_t,K+1}$, which includes the key attributes $R_h(K)$ selected for a specific XAI technique, as input for the AR extraction algorithm, serves as a means of ascertaining the true importance and degree of significance associated with these chosen attributes. This approach facilitates exploring the relationships and patterns within the data and discovering the relevance and impact of the selected attributes determined by the XAI technique. It provides a valuable avenue for evaluating the effectiveness of the attributes chosen to contribute to the interpretability and explanatory power of the XAI methodology under consideration. Afterward, the quality of the set $Q_h(K)$ has to be measured. The new metric RExQUAL based on measures for the set $Q_h(K)$ is proposed to evaluate the XAI technique's explanations.

E. Global Metrics for Association Rule Sets

Typically, ARs are individually evaluated using individual measures like confidence, lift, support, and coverage, among others [22], [23]. However, this paper proposes a different approach by evaluating the rules collectively on a global scale. Thus, new metrics specifically designed to assess the quality of a set of rules are proposed.

Let $Q_h(K)$ be a set of M rules defined as:

$$Q_h(K) = \{R_1, R_2, \ldots, R_M\}$$

where every R_i denotes a relation such as $A_i \Longrightarrow C_i$, being A_i and C_i the $i-$th antecedent and consequent, respectively, for every rule R_i.

Then, the global confidence and global support measures for a set of rules are defined.

The global support of a rule set relates to the relative frequency or proportion of transactions in the entire dataset containing all the antecedent and consequent elements specified in that particular rule set. The support is then defined as:

$$\begin{aligned}\sup(\mathbf{Q_h}(\mathbf{K})) &= \sup\left(\bigcup_{i=1}^{M}(A_i \Rightarrow C_i)\right) = \sup\left(\bigcup_{i=1}^{M} R_i\right) \\ &= \frac{1}{N_t}\#\left(\bigcup_{i=1}^{M} R_i\right) \end{aligned} \quad (11)$$

where $\#(\bigcup_{i=1}^{M} R_i)$ is the number of instances covered by the set of rules $\mathbf{Q_h}(\mathbf{K})$ in the dataset $X_h^{N_t,K+1}$, composed of N_t instances.

The cardinality of the union of the M rules is calculated as:

$$\#\left(\bigcup_{i=1}^{M} R_i\right) = \#\left(\left(\bigcup_{i=1}^{M} A_i\right)\bigcap\left(\bigcup_{i=1}^{M} C_i\right)\right) \quad (12)$$

The cardinality of the intersection of any two groups G_1 and G_2 is computed by:

$$\#\left(G_1 \bigcap G_2\right) = \#(G_1) + \#(G_2) - \#\left(G_1 \bigcup G_2\right) \quad (13)$$

Therefore applying (13) to (12) the following formula is obtained:

$$\begin{aligned}\sup(\mathbf{Q_h}(\mathbf{K})) = \frac{1}{N_t}\Bigg[&\#\left(\bigcup_{i=1}^{M} A_i\right) + \#\left(\bigcup_{i=1}^{M} C_i\right) \\ &- \#\left(\left(\bigcup_{i=1}^{M} A_i\right)\bigcup\left(\bigcup_{i=1}^{M} C_i\right)\right)\Bigg] \end{aligned} \quad (14)$$

The inclusion-exclusion principle calculates the cardinal of a union of several groups on the basis of the cardinality of every group and all possible intersections. That is,

$$\begin{aligned}\#\left(\bigcup_{i=1}^{M} G_i\right) = &\sum_{i=1}^{M}\#(G_i) - \sum_{\substack{i,j=1\\ i<j}}^{M}(G_i \cap G_j) \\ &+ \sum_{\substack{i,j,k=1\\ i<j<k}}^{M}(G_i \cap G_j \cap G_k) - \ldots \\ &+ (-1)^{M-1}\#(G_1 \cap \ldots \cap G_M) \end{aligned} \quad (15)$$

where G_i can identify either A_i or C_i.

Global confidence provides an aggregated measure of confidence across the entire set of rules and serves as an indicator of how reliable, on average, all the rules in the set are in terms of the relationship between the antecedent and the consequent.

Once defined the global support for a set of association rules, the global confidence can be extended as follows:

$$conf(\mathbf{Q_h}(\mathbf{K})) = sup(\mathbf{Q_h}(\mathbf{K}))\left(sup\left(\bigcup_{i=1}^{M} A_i\right)\right)^{-1} \quad (16)$$

where:

$$sup\left(\bigcup_{i=1}^{M} A_i\right) = \frac{1}{N_t}\#\left(\bigcup_{i=1}^{M} A_i\right) \quad (17)$$

For a better understanding of such measures, an example in which global metrics are calculated using a toy dataset is presented in the Appendix A, available online.

F. RExQUAL: The Proposed Metric

Ultimately, the new metric, RExQUAL, measures the quality of explanations based on rules. The computation of the RExQUAL measure involves utilizing the global quality measures obtained for each set of rules $Q_h(K)$. When calculating this metric, considerations include global support, global confidence, and the proportion of the number of attributes that are in the antecedent of the set of rules $Q_h(K)$ in relation to the selected key attributes $R_h(K)$. The global support metric reflects the instances in the input data that are encompassed by the set of ARs and, simultaneously, global confidence delves into the precision of the associations, reflecting the reliability of the rules in capturing meaningful patterns within the data. This dual assessment of global support and confidence helps measure how well the rules align with the underlying data distribution and the robustness of the associations between the attributes. The final XAI measure is calculated as follows:

$$RExQUAL = conf\left(Q_h(K)\right) \cdot sup\left(Q_h(K)\right) \cdot \frac{\#(F)}{K} \quad (18)$$

where F is the set of features that appear in the antecedents of rules in $Q_h(K)$. That is,

$$F = \{x_l^* \in A_i : l \in \{j+1, \ldots, j+K\} \quad A_i \in Q_h(K)\} \quad (19)$$

In this manner, the RExQUAL metric takes into account the number of attributes that are truly significant from the XAI technique and the number of features appearing in the set of ARs, as well as the degree of confidence and support provided by the set of ARs in making predictions. The RExQUAL measure varies between 0 and 1, being 1 when all attributes are present in the set of rules and the rules have global support and global confidence of 100%. The RExQUAL will approach zero when the rule set consists of very general rules with few attributes or low global support or confidence.

The computational cost of RExQUAL is directly influenced by the specific rule extraction technique applied. Once a set of rules is extracted, RExQUAL evaluates their quality by calculating global support and confidence metrics. These global metrics require processing the entire dataset, resulting in varying computational costs depending on the dataset size and complexity. As such, the overall efficiency of RExQUAL is tied to both the number of rules generated and the computational demands of calculating these global metrics.

The proposed methodology measures the quality of the explanations given by an explainability technique to the predictions obtained from any machine learning model and any type of data. The accuracy of the predictions and the computation time depend on both the size and the quality of the training dataset, whatever the machine learning model chosen. In general, the size and quality of the training data set determine the degree of generalization of a model to unseen data and the accuracy and robustness of the predictions, and the size directly influences the computation time taken to obtain the predictions. The larger the error of the predictions, the lower the quality of the explanations and, therefore, the lower the RExQUAL measure. However, the aim of the paper is not to obtain a machine learning model that obtains the best predictions but to find a measure of how good the explanations provided by an XAI technique are.

IV. Time Series Explainability: A Study Case

This section conducts experimental validation of the previously introduced methodology, focusing on evaluating the performance of various XAI techniques using the RExQUAL metric. The methodology is applied and evaluated using two distinct types of time series input data: univariate and multivariate. Detailed descriptions of these time series datasets are provided in Section IV-A. These datasets serve as input to a DL model, specifically an MLP model, as described in Section IV-B. Subsequently, in Section IV-C, the generated predictions are elucidated using the two XAI techniques, the widely recognized SHAP and LIME methods and the novel RULEx approach, described in [43]. Finally, the ARs mining method used is the Apriori algorithm, whose configuration is fully described in Section IV-D.

A. Input Data

The methodology is specifically tested using time series data but can be adapted for various data types, including tabular or image data, as it provides a flexible framework applicable to different data formats. In this case, two distinct time series datasets are input data for testing the proposed methodology.

These time series data have been chosen because they provide a diverse testing ground for the methodology. One dataset is univariate, offering a simpler structure to evaluate the interpretability techniques, while the multivariate dataset introduces complexity, allowing for the assessment of how the XAI techniques handle interdependencies between multiple variables. Additionally, DL models have already been trained and tested on these datasets [42], [43], further justifying their selection.

On the one hand, the first dataset consists of a univariate time series that captures the demand for electrical energy, with measurements expressed in kilowatt-hours (kWh) in Spain [38]. These data span nine years and six months, from January 1st, 2007, to June 21st, 2016. Measurements are recorded with a 10-minute frequency. The input comprises a window of 168 past values (1 d and 4 hours) utilized to predict a subsequent prediction horizon of 24, representing a 4-hour interval. The input data consists of 10161 instances, with 192 time stamps (1 d and 8 hours). On the other hand, the multivariate time

series data comprise multiple meteorological variables from the *Sistema Agrometeorológico para a Gestão da Rega no Alentejo*, in Portugal (SAGRA). The data is recorded by sensors from eight stations in the Alentejo region and covers the period from August 2012 to 2022. The different variables are recorded with a daily frequency, resulting in 29350 instances. This data set has previously been utilized in various research papers dedicated to time series forecasting [17], [42]. Fourteen weather-related variables were collected, namely mean temperature, maximum temperature, minimum temperature (°C), mean relative humidity, maximum relative humidity, minimum relative humidity (%), global solar radiation (kj/m^2), wind direction (graus), mean wind velocity, maximum wind velocity (m/s), precipitation (mm), mean soil temperature, maximum soil temperature and minimum soil temperature (°C). These variables were used to estimate reference evapotranspiration, denoted ET0, the target variable. The time series input data, featuring a window size of 3 and a prediction horizon of 7 days, include ET0 values and other meteorological features.

The time series data were preprocessed for analysis before applying DL techniques. For the first dataset, missing values were filled using linear interpolation, followed by a windowing technique with 168 input elements (1 d and 4 hours) and 24 output elements (4-hour forecast). For the second dataset, data from multiple stations were unified, and missing values were imputed using a k-nearest neighbors algorithm. A windowing method was then applied, with 3 input elements (3 days) and 7 output elements (7 days). Finally, the datasets were split into training (70%) and testing (30%) subsets.

B. Deep Learning: MLP

In this step, an MLP model is trained to generate predictions. An MLP model has been selected due to its standardized architecture, availability in various open-source libraries, and proven effectiveness for diverse forecasting tasks. Specifically, the MLP utilized in this study consists of three hidden layers, each with 128, 64, and 32 units, which employ the ReLU activation function. The output layer is designed with the same number of units as required for the specific prediction task. In the context of time series prediction, there are H units aligned to predict H events within the prediction horizon. The loss function is the mean squared error and the model is trained through 50 epochs.

C. XAI

Various post-hoc or model-independent XAI techniques can be used in this methodology phase. Specifically, two distinct approaches are evaluated in this section: SHAP, RULEx and LIME. In addition, a random selection technique is used as a reference point to assess the performance of XAI methods. This random method randomly picks a subset of significant features from the dataset, providing a baseline for comparison. To begin with, SHAP is a powerful framework grounded in cooperative game theory that allocates the contribution of each feature to the prediction of an ML model. In this way, at the core of SHAP lies the Shapley value, a concept introduced in [35] by Lloyd Shapley that assigns a unique contribution to each player in a coalition game, reflecting their marginal impact on different possible collaborations. In the context of ML, each feature represents a "player," and the prediction outcome is analogous to the value generated by the coalition of all features. In contrast, RULEx, recently presented in [43], [44], aims to improve the interpretability of time series forecasting through visual representations of sets of rules. The set of ARs captures relationships among variables. The ARs set serves as the input for a graphic explanation process, which is a model-agnostic explainability technique. This means that RULEx is elucidating the workings of the model by illustrating the most important values for the time series forecasting task. The output is a graph-based representation, colored to reflect the importance of the input values in influencing the predictions. This importance is computed for each distinct value of the prediction horizon. Later, LIME is based on generating locally interpretable explanations for individual predictions. LIME approximates the behavior of a complex model by fitting a simpler, interpretable model, such as linear regression, to perturbations of the input data around the instance being explained. In this work, LIME is executed a fixed number of times on random data samples to generate a global representation of its results. Finally, a random feature selection was performed as a baseline reference. Approximately 15% to 25% of all attributes in each data set were randomly chosen. Subsequently, this random selection was treated as the attribute ranking process described above.

D. XAI Evaluation: Apriori Algorithm

The Apriori ARs mining method is used to obtain a set of ARs. The ARs are then evaluated using the overarching metrics detailed in Section III-E. Finally, the XAI technique is evaluated by the RExQUAL indicator computed as described in Section III-F.

The Apriori algorithm is chosen for its ability to uncover associations between items and its ease of application through Python libraries such as $mlxtend$ [31]. Additionally, the Apriori algorithm initiates by scanning the dataset to calculate the support of each element, and elements exceeding a predefined minimum support threshold are identified as frequent. Various threshold values are tested in this experiment, specifically 0.1, 0.15, and 0.2. Furthermore, Apriori is designed to work with categorical or binary data. Converting them into discrete categories is essential for the Apriori algorithm to be applied effectively in the context of continuous variables. In addition, discretization involves dividing the continuous range of values into intervals or bins and assigning each data point to a specific interval. In this experiment, various discretization bins were evaluated to determine the optimal number of intervals for continuous input data. Specifically, the values tested were 3, 5, and 7 bins, uniformly sized based on the range of the feature. The selection of these specific numbers is based on a human-understandable division: 3 corresponds to low-normal-high, 5 encompasses very low-low-normal-high-very high, and 7 delineates very low-medium low-few low-normal-few high-medium high-very high categories.

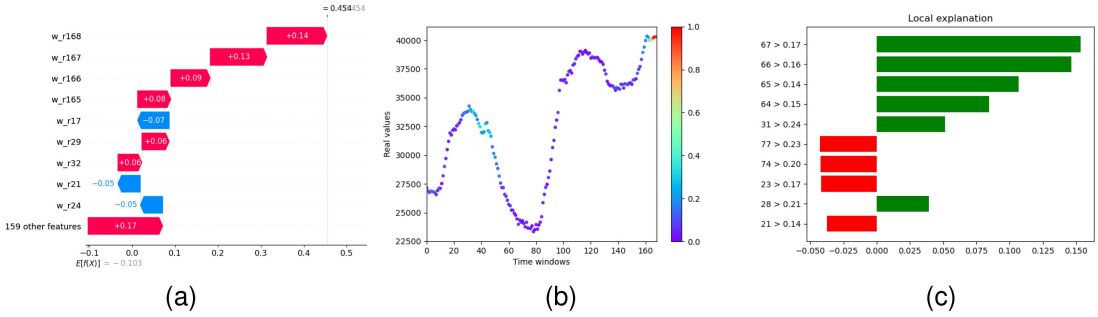


Fig. 2. Explanations generated for a random instance of the electric demand dataset (time horizon $h = 2$). (a) SHAP (b) RULEx (c) LIME.

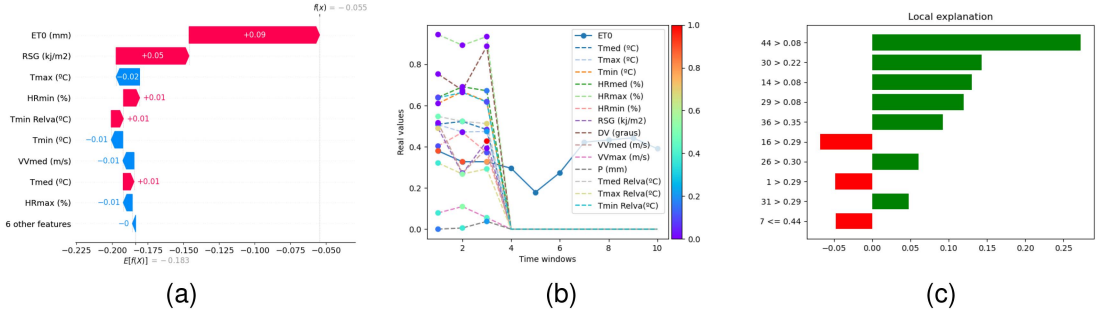


Fig. 3. Explanations generated for a random instance of the SAGRA dataset (time horizon $h = 4$). (a) SHAP (b) RULEx (c) LIME.

TABLE II
ASSOCIATION BETWEEN THE x NUMERICAL VALUES AND THE SAGRA DATASET FEATURES

Feature	x index		
	t-3	t-2	t-1
Tmed	0	15	30
Tmax	1	16	31
Tmin	2	17	32
HRmed	3	18	33
HRmax	4	19	34
HRmin	5	20	35
RSG	6	21	36
DV	7	22	37
VVmed	8	23	38
VVmax	9	24	39
P	10	25	40
Tmed Relva	11	26	41
Tmax Relva	12	27	42
Tmin Relva	13	28	43
ET0	14	29	44

V. RESULTS

This section presents the results of the experiment carried out to test the proposed methodology, emphasizing the outputs generated by the XAI techniques, SHAP, RULEx and LIME. Lately, the proposed metric RExQUAL is calculated. All XAI techniques are evaluated using RExQUAL and compared against a baseline consisting of randomly selected features.

A. XAI

First, it is important to consider Table II, which encodes the numerical indices of the SAGRA features, as the various features of the SAGRA dataset are presented consecutively as time indices. Table II clarifies the results presented in this section by linking the features at various time points to the corresponding numbers. This Table is useful for the Y-axis of the LIME output in Fig. 3(c) and the X-axis of the rankings in Fig. 5. For example, the attribute at positions 0, 15, and 30 corresponds to the same feature, $Tmed$, across the three-time window steps.

The predictions made by the MLP model for the two input datasets are explained using SHAP, RULEx and LIME. The results of each of these techniques are described below. Random local examples of each XAI technique outputs are presented in Figs. 2 and 3. These Figures show the level of importance of the attributes for a specific instance of the data.

In the context of electric demand, the results of the XAI techniques are presented for the predictions made for the time horizon $h = 2$ in Fig. 2. In Fig. 2(a), the SHAP plot offers detailed information on crucial features and their impact on specific predictions through visual representation. Meanwhile, the RULEx diagram in Fig. 2(b) uses colorization to denote the importance levels of each characteristic in representing the time series values. In both cases, the predominant key features are the most recent ones, that is, the latest values from the time window, specifically spanning the time window from 165 to 168. This observation implies that the latest electric demand measurements are pivotal in predicting demand for subsequent time steps. Furthermore, specific values of time series corresponding to time windows such as w_{17}, w_{29} or w_{32} (in Fig. 2(a)) and roughly between w_{25} and w_{50} (in Fig. 2(b)) also have a certain level of importance in prediction generation. These values signify the 24 hours preceding the predicted value. In the

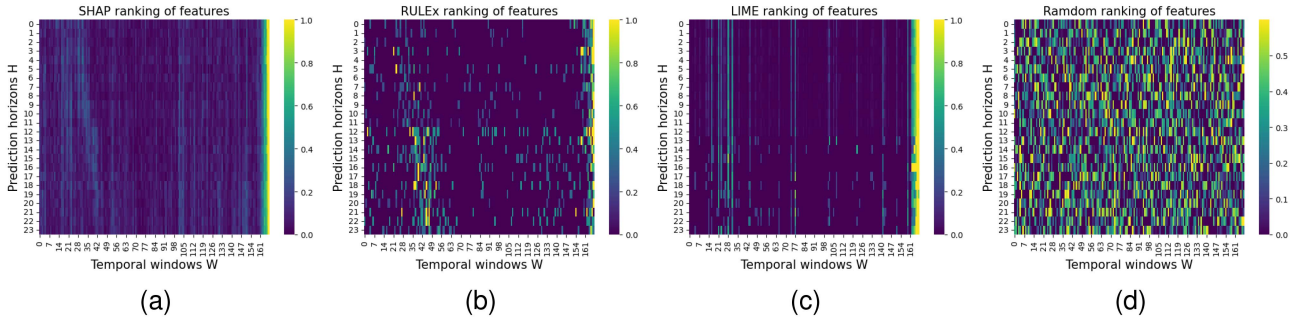


Fig. 4. Feature importance for each input time window and for the features in the electric demand dataset. (a) SHAP (b) RULEx (c) LIME (d) Random.

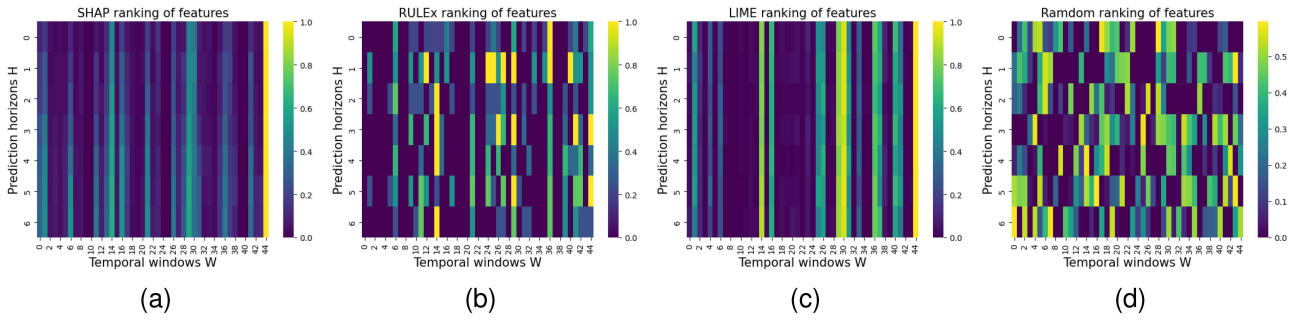


Fig. 5. Feature importance values for each input time window and for the features in the SAGRA dataset. (a) SHAP (b) RULEx (c) LIME (d) Random.

case of LIME, in Fig. 2(c) the highlighted important values are ranging between $w_2 1$ and w_{67}, showing any discernible patterns.

Concerning SAGRA, Fig. 3 presents a visualization of the explanations obtained by the XAI techniques for the predictions of a random instance for the time horizon $h = 4$. The SHAP diagram (Fig. 3(a)) exclusively displays the variable names without accounting for the time windows. The three most influential variables include the target value itself, ET0 (mm), solar radiation (RSG in kg/m^2), and maximum air temperature (Tmax). In the case of the RULEx output (Fig. 3(b)), the graph includes the representation of the target variable, ET0, and the significance values associated with the prominent attributes, such as RSG, or ET0 itself. The fluctuation of the target variable, ET0, is denoted by the blue line, capturing the temporal dynamics within both the preceding time window (from $t-3$ to $t-1$) and the predicted values (from t to $t+6$). The remaining variables are only depicted within the specified time window, serving as input, as indicated in the legend. Values are normalized to ensure uniform representation across the graph. Subsequently, the colored points in the graph represent the importance values of the input data from $t-3$ to $t-1$. The significance of each variable at each time point is illustrated by the color gradient, with red indicating the highest level of importance and purple indicating the lowest. In this specific sample, the color of the points transitioning from yellow to red for RSG and red for ET0 indicate that RSG and ET0 are crucial features, aligning with the results obtained from SHAP. The significance of the features remains consistent throughout the time intervals from $t-3$ to $t-1$, with a notable emphasis on the most recent moment $t-1$. Finally, in the LIME output shown in Fig. 3(c), three of the most important features (14, 29, and 44) correspond to past values of ET_0, as observed with the other XAI techniques. Additionally, other important features are related to meteorological data from the most recent time period, including mean temperature (30), maximum temperature (31), and RSG (36).

However, from a global perspective, the current emphasis is shifting toward the rankings $R_h(K)$ acquired for each time horizon h and the corresponding identification of K key attributes. The rankings $R_h(K)$ are visually presented by heatmaps for the electric demand dataset in Fig. 4 and for the SAGRA data in Fig. 5. The color code highlights the importance of the attributes, with values near 1 indicating significant relevance for a specific prediction horizon h, while those near 0 suggest lesser importance. Regarding the rankings of electric demand, Fig. 4 illustrates a similar distribution of the importance values of the features identified by SHAP and RULEx. Notably, key features are prominently clustered in the dataset's later phase, roughly 160 to 168, underscoring the critical role of recent events in predictions. Furthermore, a discernible group of influential features exists around w_{24} at $h = 0$ and persists over time horizons, reaching approximately w_{48} at $h = 23$. Considering that 144 values correspond to 24 hours, given the 10-minute frequency of the data, this pattern suggests that events occurring 24 hours earlier also bear significance in the model predictions. Significantly, this scheme is more pronounced in the results obtained by RULEx (Fig. 4(b), particularly from $h = 12$ onwards. Additionally, concerning the LIME ranking, in Fig. 4(c), the moving pattern around the 24 hours prior is not observed.

TABLE III
NUMBER OF ARS FOR $h = 2$ IN ELECTRIC DEMAND DATASET

XAI	Min sup	Number of bins		
		3	5	7
SHAP	0.1	∞	3401	446
	0.15	35877	**334**	42
	0.2	2099	95	0
RULEx	0.1	∞	645	227
	0.15	15307	**152**	41
	0.2	1065	63	0

TABLE IV
NUMBER OF ARS FOR $h = 4$ IN SAGRA DATASET

XAI	Min sup	Number of bins		
		3	5	7
SHAP	0.1	29793	1833	11
	0.15	4026	**137**	3
	0.2	780	12	0
RULEx	0.1	490	192	39
	0.15	318	**48**	15
	0.2	270	21	3

Instead, there is a straight line between w_{14} and w_{28}, which roughly corresponds to 24 hours earlier, but it does not change as the prediction horizon progresses. Additionally, there are points of relative importance around w_{77} and w_{140}.

Regarding the SAGRA data rankings shown in Fig. 5, it is worth noting that due to its multivariate nature, the 15 attributes are replicated three times. The importance of features is uniformly distributed across the three distinct time windows. There are no notable differences in the patterns for each prediction horizon. In the case of SHAP (Fig. 5(a)), a distinct pattern emerges, highlighting the ET0 attribute as the most crucial feature (positions 14, 29, and 44), particularly underscoring the significance of the most recent value (position 44). On the contrary, when looking at RULEx (Fig. 5(b)), there is a wider spectrum of attributes identified as important, extending beyond ET0. For example, RSG (in positions 6, 21, and 36) demonstrates notable importance. This fact has been corroborated by previous evidence indicating a significant relationship with ET0 forecasting [42]. Additionally, regarding the LIME output (Fig. 5(c)), the past values of ET_0 are clearly highlighted (yellow lines at points 14, 30, and 44). In general, the most important features are the most recent ones (from points 30 to 44), regardless of the type of meteorological data. Finally, the random selection of key attributes (Figs. 4 and 5(c)) lack any discernible pattern for both datasets, as evidenced by its inherently unpredictable nature.

B. Apriori Configuration

Once the key features $R_h(K)$ are identified for each value h of the prediction horizon, the initial dataset undergoes refinement, retaining only the K features identified as significant. Subsequently, the refined dataset is employed as input for ARs generation using Apriori.

As introduced in Section IV-D, the results obtained by Apriori depend significantly on the number of bins into which the input data are divided and the minimum support set as a threshold. To determine the optimal configuration of these parameters, an experimental setup is carried out for each value of the prediction horizon. ARs are extracted with values for the number of bins and minimum support varying by $\{3, 5, 7\}$ and $\{0.1, 0.15, 0.2\}$, respectively. The number of rules generated by Apriori for a random prediction horizon for each input data (namely $h = 2$ for the electric demand data and $h = 4$ for the SAGRA data) and for the key attributes of each of the XAI techniques are presented in Tables III and IV, respectively. In both cases, whether using SHAP or RULEx, and employing 3 bins, rules are abundant. However, when using 7 bins, there is a scarcity of rules and, in certain cases, none. This observed trend suggests that fewer bins generate many rules and lose importance. In addition, excessive data subdivisions into bins may result in the loss of association patterns. Then, the configuration of 5 bins ends in a manageable number of rules with a reasonable compromise between rule quantity and significance.

Subsequently, the objective is to identify the optimal threshold for minimum support. For that reason, the global measures of each of the rule sets obtained by Apriori with 5 bins are presented in Tables V (for the electrical demand data) and VI (for the SAGRA data). These tables show the mean and standard deviation of the global measures for all values h of the prediction horizons for the 5-bin configuration. The average of the support and confidence of the ARs are included to demonstrate that the global metrics do not equal the average of the individual metrics. The tables show that the best setup for minimum support is 0.15, resulting in the most balanced and meaningful rules for both SHAP and RULEx. A minimum support of 0.1 yields satisfactory results, but it produces an excessive number of rules. On the other hand, securing ARs with minimum support of 0.2 is overly limiting and diminishes meaningfulness, as seen in the reduced values of global support and confidence. At the end, following this experimental process, the optimal setup for Apriori is identified as having a minimum support of 0.15 and a discretization of 5 bins. To reduce computational costs, the Apriori algorithm was configured to implement rule pruning methods that discard redundant rules, focusing on the most informative ones. Additionally, parallel computing techniques can be applied to distribute the rule generation process across multiple cores, improving overall efficiency.

C. RExQUAL

Finally, following the extraction of ARs, the subsequent phase involves the computation of global measures of the rule sets and then culminates in the calculation of RExQUAL to measure the quality of the explanations obtained by XAI techniques. The RExQUAL calculation utilizes global measures of the rule sets, specifically support and confidence, as outlined in Section III-D. RExQUAL also considers the number of crucial attributes selected by the XAI methods (K) and how many of these characteristics are included in the antecedent of the rule set $Q_h(K)$. All these values and the final RExQUAL metric for the XAI techniques are presented in Tables VII and VIII for electricity demand and SAGRA datasets, respectively. In addition, the RExQUAL measure for each value of the prediction horizon is presented in Fig. 6 for the two datasets used in the

TABLE V
GLOBAL METRICS FOR THE RULE SET OBTAINED BY APRIORI WITH 5 BINS USING THE KEY ATTRIBUTES SELECTED IN THE ELECTRIC DEMAND DATASET

XAI	Min Sup	#Attributes	#Rules	Mean Support	Mean Conf	Global Sup	Global Conf
SHAP	0.1	27.7 ± 5	3558 ± 9828	0.117 ± 0.005	0.695 ± 0.124	0.606 ± 0.180	0.705 ±0.130
	0.15	27.7 ± 5	191 ± 295	0.171 ± 0.009	0.629 ± 0.009	**0.606 ± 0.180**	**0.705 ± 0.130**
	0.2	27.7 ± 5	27 ± 65	0.152 ± 0.100	0.494 ± 0.329	0.258 ± 0.209	0.481 ± 0.246
RULEx	0.1	24.3 ± 6.9	1293 ± 1253	0.122 ± 0.005	0.699 ± 0.096	0.870 ± 0.066	0.872 ±0.067
	0.15	24.3 ± 6.9	142 ± 159	0.173 ± 0.010	0.654 ± 0.076	**0.626 ± 0.129**	**0.702 ± 0.119**
	0.2	24.3 ± 6.9	26 ± 63	0.204 ± 0.048	0.657 ± 0.167	0.331 ± 0.158	0.618 ± 0.165

TABLE VI
GLOBAL METRICS FOR THE RULE SET OBTAINED BY APRIORI WITH 5 BINS USING THE KEY ATTRIBUTES SELECTED BY IN SAGRA DATASET

XAI	Min Sup	#Attributes	#Rules	Mean Sup	Mean Conf	Global Sup	Global Conf
SHAP	0.1	15.9 ± 2.8	1189 ± 608	0.122 ± 0.002	0.711 ± 0.020	0.797 ± 0.004	0.798 ± 0.003
	0.15	15.9 ± 2.8	108 ± 28	0.172 ± 0.003	0.678 ± 0.035	**0.479 ± 0.149**	**0.678 ± 0.155**
	0.2	15.9 ± 2.8	11 ± 1	0.215 ± 0.002	0.640 ± 0.045	0.349 ± 0.163	0.573 ± 0.127
RULEx	0.1	7.6 ± 1.6	193 ± 126	0.245 ± 0.008	0.659 ± 0.511	0.944 ± 1.011	0.951 ± 0.073
	0.15	7.6 ± 1.6	60 ± 41	0.287 ± 0.011	0.663 ± 0.067	**0.910 ± 0.180**	**0.948 ± 0.081**
	0.2	7.6 ± 1.6	23 ± 20	0.298 ± 0.009	0.616 ± 0.087	0.730 ± 0.206	0.791 ± 0.091

TABLE VII
GLOBAL METRICS AND REXQUAL CALCULATIONS FOR ELECTRIC DEMAND DATASET

	SHAP					RULEx					LIME					Random				
h	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL
0	0.819	0.891	27	13	0.351	0.820	0.899	20	10	0.368	0.407	0.525	16	12	0.160	0.783	0.875	30	4	0.091
1	0.790	0.864	16	10	0.427	0.820	0.899	20	10	0.368	0.388	0.559	18	12	0.145	0.783	0.875	30	4	0.091
2	0.802	0.871	19	12	0.441	0.797	0.870	13	8	0.427	0.423	0.544	19	10	0.121	0.783	0.862	33	7	0.143
3	0.791	0.859	25	13	0.354	0.750	0.821	19	10	0.324	0.410	0.528	16	10	0.135	0.768	0.847	32	7	0.142
4	0.786	0.852	27	12	0.298	0.777	0.850	11	8	0.480	0.393	0.511	17	11	0.130	0.776	0.847	34	7	0.135
5	0.779	0.844	21	11	0.343	0.778	0.842	28	13	0.304	0.444	0.489	15	9	0.130	0.762	0.833	41	5	0.077
6	0.774	0.837	30	14	0.302	0.757	0.832	13	6	0.291	0.285	0.405	18	12	0.077	0.751	0.809	32	6	0.114
7	0.761	0.825	24	12	0.314	0.766	0.813	29	14	0.301	0.281	0.498	16	12	0.105	0.686	0.742	35	5	0.073
8	0.740	0.804	24	12	0.297	0.687	0.726	19	9	0.236	0.273	0.489	17	13	0.110	0.600	0.657	25	7	0.110
9	0.679	0.734	34	14	0.205	0.676	0.721	18	11	0.298	0.272	0.491	15	9	0.080	0.413	0.592	24	3	0.031
10	0.496	0.589	31	8	0.075	0.494	0.582	22	10	0.131	0.361	0.419	17	10	0.089	0.481	0.578	29	7	0.067
11	0.485	0.580	28	8	0.080	0.470	0.563	28	11	0.104	0.354	0.415	17	11	0.095	0.447	0.643	39	3	0.022
12	0.700	0.744	29	12	0.216	0.703	0.751	38	8	0.111	0.261	0.313	15	10	0.054	0.613	0.705	31	2	0.028
13	0.654	0.712	25	14	0.261	0.494	0.576	23	6	0.074	0.286	0.461	6	4	0.088	0.657	0.701	34	8	0.108
14	0.511	0.602	33	6	0.056	0.525	0.560	31	8	0.081	0.279	0.342	10	7	0.067	0.493	0.593	33	7	0.062
15	0.506	0.607	26	4	0.047	0.520	0.596	32	6	0.058	0.274	0.433	9	5	0.066	0.507	0.582	39	12	0.091
16	0.526	0.614	31	5	0.052	0.533	0.603	33	7	0.068	0.271	0.372	8	6	0.075	0.545	0.593	50	7	0.045
17	0.244	0.672	27	4	0.024	0.535	0.606	18	4	0.072	0.272	0.443	8	6	0.091	0.518	0.563	36	4	0.032
18	0.573	0.616	39	6	0.054	0.562	0.620	25	7	0.098	0.167	0.206	11	9	0.028	0.552	0.617	40	9	0.077
19	0.523	0.606	27	3	0.035	0.550	0.625	27	8	0.102	0.163	0.261	10	7	0.030	0.541	0.604	39	6	0.050
20	0.522	0.603	30	3	0.032	0.542	0.601	29	6	0.067	0.155	0.362	8	6	0.042	0.542	0.640	41	3	0.025
21	0.472	0.597	34	3	0.025	0.509	0.618	27	5	0.058	0.159	0.293	11	7	0.030	0.510	0.640	45	3	0.023
22	0.451	0.574	31	1	0.008	0.468	0.650	29	7	0.073	0.140	0.227	5	4	0.025	0.506	0.635	41	3	0.024
23	0.153	0.428	27	1	0.003	0.500	0.628	29	6	0.064	0.123	0.137	9	7	0.013	0.461	0.583	44	3	0.018
Mean	0.606	0.705	27.7	8.4	**0.179**	0.626	0.702	24.3	8.3	**0.191**	0.283	0.405	13.1	8.7	**0.083**	0.603	0.692	36.1	5.7	**0.073**
Dev.	0.180	0.130	5.0	4.5	0.151	0.129	0.119	6.9	2.6	0.129	0.096	0.116	4.4	2.7	0.041	0.127	0.111	6.3	2.5	0.044

TABLE VIII
GLOBAL METRICS AND REXQUAL CALCULATIONS FOR SAGRA DATASET.

	SHAP					RULEx					LIME					Random				
h	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL
0	0.719	0.861	11	9	0.130	0.978	0.979	7	7	0.956	0.235	0.235	13	11	0.047	0.978	0.978	12	6	0.479
1	0.555	0.797	15	14	0.413	0.979	0.979	7	7	0.959	0.234	0.234	13	11	0.046	0.449	0.722	16	10	0.203
2	0.524	0.754	14	13	0.367	0.502	0.764	5	5	0.384	0.230	0.230	15	13	0.046	0.293	0.492	9	6	0.096
3	0.497	0.726	16	15	0.338	0.978	0.979	10	10	0.958	0.227	0.227	13	10	0.040	0.979	0.979	16	12	0.719
4	0.448	0.668	18	16	0.266	0.979	0.979	8	8	0.958	0.215	0.215	13	11	0.039	0.979	0.979	11	7	0.610
5	0.296	0.469	19	16	0.117	0.978	0.979	9	9	0.958	0.215	0.215	13	9	0.032	0.978	0.979	17	12	0.676
6	0.295	0.468	18	16	0.123	0.978	0.979	7	7	0.958	0.214	0.214	13	9	0.032	0.979	0.979	13	8	0.590
Mean	0.479	0.678	15.9	14.1	**0.251**	0.910	0.948	7.6	7.6	**0.876**	0.224	0.224	13.3	10.6	**0.040**	0.805	0.873	13.4	8.7	**0.482**
Dev.	0.149	0.155	2.8	2.5	0.127	0.180	0.081	1.6	1.6	0.217	0.009	0009	0.7	1.3	0.006	0.300	0.193	3.0	2.6	0.241

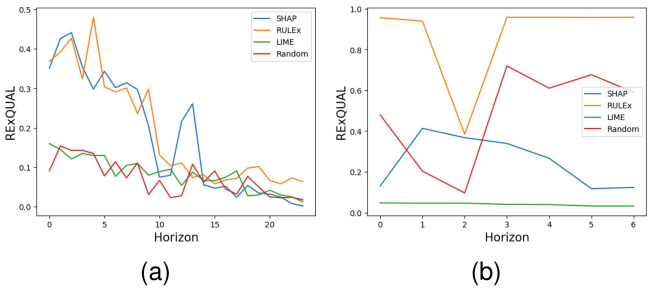


Fig. 6. REXQUAL metric across the prediction horizons. (a) Electric demand dataset (b) SAGRA dataset.

experiment. The visual representation simplifies the comparison between the two XAI techniques and the random benchmark. Additionally, a stability analysis of REXQUAL was conducted to evaluate its robustness under small data perturbations, confirming its reliability across scenarios. Due to space constraints, the results are available in the project's GitHub repository.[2]

First, from Table VII, it can be noticed that RULEx emerges as the superior XAI method according to REXQUAL, boasting an average of 0.191 ± 0.129. SHAP, while not far behind, records a mean value of 0.179 ± 0.151. Predictably, the selection of random features is shown to be the least effective choice, with an average of 0.073 ± 0.044. SHAP and RULEx results obtain similar values of global support, confidence, and key features. It should be noted that, over time, as the horizon progresses (h), the number of key attributes chosen by the XAI technique (K) remains constant, while the number of attributes contained within the antecedent of the rules ($Q_h(K)$) decreases as time passes. In

[2][Online]. Available: https://github.com/Data-Science-Big-Data-Research-Lab/rexqual.

TABLE IX
SUMMARY OF THE REXQUAL CALCULATION METRICS, TO COMPARE SHAP, RULEX AND LIME FOR BOTH DATASETS

Dataset	XAI	K	$Q_h(K)$	#Rules	Mean Sup	Mean Conf	Global Sup	Global Conf	RExQUAL
Electric	SHAP	27.7 ± 5	8.4 ± 4.5	3558 ± 9828	0.117 ± 0.005	0.695 ± 0.124	0.867 ± 0.069	0.868 ±0.069	0.179 ± 0.151
Electric	RULEx	24.3 ± 6.9	8.3 ± 2.6	142 ± 159	0.173 ± 0.010	0.654 ± 0.076	0.626 ± 0.129	0.703 ± 0.117	**0.191 ± 0.129**
Electric	LIME	13.1 ± 4.4	8.7 ± 2.7	163 ± 186	0.123 ± 0.008	0.589 ± 214	0.283 ± 0.096	0.405 ± 0.116	0.083 ± 0.041
SAGRA	SHAP	15.9 ± 2.8	14.1 ± 2.5	108 ± 28	0.172 ± 0.003	0.678 ± 0.035	0.416 ± 0.117	0.632 ± 0.138	0.251 ± 0.127
SAGRA	RULEx	7.6 ± 1.6	7.6 ± 1.6	60 ± 41	0.287 ± 0.011	0.663 ± 0.067	0.882 ± 0.180	0.948 ± 0.081	**0.876 ± 0.217**
SAGRA	LIME	13.3 ± 0.7	10.6 ± 1.3	370.0 ± 73	0.344 ± 0.016	0.389 ± 0.171	0.224 ± 0.009	0.224 ± 0.009	0.040 ± 0.006

contrast, LIME demonstrates significantly worse performance, with an average RExQUAL value of 0.083 ± 0.041, but shows greater consistency in the number of selected features.

From Table VIII, for SAGRA dataset, the quality of the explanations given by RULEx is much superior, with an average of 0.876 ± 0.217. SHAP, on the other hand, does not perform well in this scenario, recording an average of 0.251 ± 0.127. Interestingly, the random feature selection method surpasses SHAP, averaging 0.482 ± 0.241. LIME remains the least effective XAI technique, achieving an average RExQUAL value of 0.040 ± 0.006. This could stem from the significant contrast in the global metrics of the AR sets, where RULEx demonstrates superior performance, boasting an average of 0.910 for global support and 0.948 for global confidence. Furthermore, this phenomenon could also be attributed to the correlation between the number of key attributes K and their occurrence in the antecedent of the rules. The average value of K is notably lower in the case of RULEx (with a mean of 7.6 ± 1.6) compared to SHAP (15.9 ± 2.8) and LIME (13.3 ± 0.7) and similar to the random baseline (13.4 ± 3.0). Regardless, for both datasets, it is noted that the global metrics of the rule sets experience a decline over time, exhibiting significant reductions more noticeable in the case of the SHAP.

Furthermore, the values of RExQUAL across time horizons are examined in Fig. 6 for both datasets. In the case of electricity demand, where there is a higher number of time horizons (namely, 24), the measure declines as the time horizon expands for the XAI techniques. The RExQUAL values for the random feature selection remain nearly stable. The quality of explanations provided by SHAP and RULEx surpasses those obtained using the random method, whereas LIME's performance is quite similar. However, for h greater than 15, their values become comparable, obtaining RULEx always the highest value for RExQUAL. This suggests that as the forecast horizon lengthens, the prediction error broadens, simultaneously reducing the clarity of the explanations provided by XAI techniques. In contrast, no discernible patterns are observed for the SAGRA dataset due to the reduced number of time horizons (namely, 7). The RExQUAL metric for RULEx significantly outperforms other approaches. Nevertheless, at certain time horizons, such as $h = 2$, notably low RExQUAL values for RULEx are observed. This could be attributed to the reduced number of key attributes selected at this particular value of time horizon, which is 5 compared to the mean of 7.6. The stochastic nature of the Apriori algorithm, which is tasked with generating association rules, could also be a source of origin. SHAP and the random approach exhibit an irregular distribution of the RExQUAL metric on time horizons. This fact is expected for the random selection of attributes, but in the case of SHAP, it suggests that its explanations are of low quality for this scenario.

Moreover, Table IX compiles all the metrics, both the average of the individual metrics for each rule and the global metrics and the proposed RExQUAL measure for the rules obtained with the proposed methodology. These metrics measure the quality of the explanations provided by RULEx, SHAP and LIME as XAI techniques for the two datasets. Note that these metrics are the mean and standard deviation of the measurements over all time horizons. Based on these findings, it can be inferred that the RULEx method achieves superior quality in its explanations for both univariate and multivariate time series, as per the proposed RExQUAL metric. In contrast, SHAP is not able to explain the predictions for the multivariate time series. LIME achieves very poor performance. In this case, it can be seen that although the mean confidence of each rule is similar for SHAP and RULEx, the global confidence for RULEx is much higher than for SHAP. It can be concluded that the proposed global measures for rule sets evaluate a set of rules more accurately than simply averaging the individual measures of each rule.

Finally, a Bayesian analysis of the contrast hypothesis is performed using the approach proposed by Corani et al. [9]. This Bayesian adaptation of the Wilcoxon signed-rank test was applied to determine the most effective XAI method according to the RExQUAL metric. The paired statistical test was applied with a significance level of 0.01 to compute the probability that one XAI method performs better than the other. The results indicate with a probability of 1 that both SHAP and RULEx outperform LIME, while the probability that RULEx is better than SHAP is 0.996. Regarding the Random method, SHAP, RULEx, and LIME each have a probability of 1 being superior.

D. Test Set

Although the primary objective of this work is to explore the inner mechanisms of DL models, the same experiments were conducted using the test set of both datasets. It is important to note that the focus is not on optimizing the model for the forecasting task; therefore, the MLP model employed here exhibits relatively poor performance compared to other models reported in the literature [12], [42]. Specifically, the model achieves a Mean Absolute Error (MAE) of 963.35 for the electricity demand dataset, whereas in the literature, the MAE is 529.57. In the case of the SAGRA dataset, the MAE is 0.841 across the seven prediction horizons, which is notably higher than the MAE of 0.765 reported for the optimized model in the literature. Tables X and XI present the global metrics required to compute RExQUAL, along with the corresponding index values, for each of the datasets. In general, the metrics show significantly

TABLE X
GLOBAL METRICS AND REXQUAL CALCULATIONS FOR ELECTRIC DEMAND DATASET USING THE TEST SAMPLE OF DATA

	SHAP					RULEx					LIME				
h	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL
0	0.408	0.515	27	12	0.093	0.490	0.502	20	8	0.098	0.456	0.467	16	9	0.120
1	0.388	0.522	16	8	0.101	0.388	0.514	20	8	0.080	0.446	0.416	18	10	0.103
2	0.376	0.508	19	7	0.070	0.354	0.506	13	6	0.083	0.436	0.445	19	8	0.082
3	0.358	0.479	25	10	0.069	0.426	0.555	19	7	0.087	0.394	0.445	16	7	0.077
4	0.413	0.442	27	9	0.061	0.369	0.492	11	6	0.099	0.410	0.405	17	9	0.088
5	0.316	0.481	21	12	0.087	0.361	0.482	28	10	0.062	0.462	0.400	15	7	0.086
6	0.356	0.482	30	12	0.069	0.373	0.501	13	6	0.086	0.420	0.436	18	8	0.081
7	0.372	0.494	24	9	0.069	0.370	0.492	29	9	0.056	0.163	0.293	16	5	0.015
8	0.367	0.512	24	10	0.078	0.366	0.463	19	5	0.045	0.216	0.316	17	7	0.028
9	0.365	0.441	34	14	0.066	0.327	0.418	18	4	0.031	0.188	0.300	15	6	0.023
10	0.379	0.422	31	12	0.062	0.273	0.418	22	5	0.026	0.120	0.289	17	6	0.020
11	0.370	0.395	28	10	0.052	0.360	0.429	28	4	0.022	0.181	0.275	17	6	0.0176
12	0.267	0.385	29	7	0.025	0.363	0.391	38	7	0.026	0.170	0.248	15	4	0.011
13	0.262	0.383	25	6	0.024	0.337	0.385	23	5	0.028	0.216	0.350	6	4	0.051
14	0.266	0.288	33	7	0.016	0.265	0.381	31	6	0.020	0.163	0.276	10	4	0.0181
15	0.265	0.380	26	7	0.027	0.266	0.388	32	6	0.019	0.180	0.286	9	5	0.029
16	0.262	0.332	31	7	0.020	0.281	0.401	33	7	0.024	0.162	0.257	8	4	0.021
17	0.284	0.321	27	7	0.024	0.295	0.412	18	4	0.027	0.176	0.235	8	4	0.021
18	0.268	0.312	39	7	0.015	0.313	0.512	25	4	0.026	0.154	0.223	11	6	0.019
19	0.221	0.312	27	6	0.015	0.322	0.398	27	4	0.019	0.148	0.227	10	5	0.017
20	0.205	0.508	30	6	0,021	0.270	0.385	29	4	0.014	0.174	0.251	8	4	0.022
21	0.187	0.542	34	6	0.018	0.287	0.497	27	3	0.016	0.154	0.225	11	6	0.019
22	0.245	0.350	31	4	0.011	0.293	0.402	29	3	0.012	0.143	0.205	5	3	0.018
23	0.160	0.481	27	6	0.017	0.300	0.509	29	2	0.011	0.157	0.216	9	4	0.015
Mean	0.307	0.429	27.7	8.4	**0.046**	0.335	0.451	24.3	5.5	**0.042**	0.245	0.312	13.1	5.9	**0.042**
Dev.	0.072	0.078	5.0	2.5	0.029	0.054	0.054	6.9	2.0	0.030	0.122	0.084	4.4	1.9	0.033

TABLE XI
GLOBAL METRICS AND REXQUAL CALCULATIONS FOR SAGRA DATASET USING THE TEST SAMPLE OF DATA

	SHAP					RULEx					LIME				
h	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL	G Sup	G Conf	K	$Q_h(K)$	RExQUAL
0	0.235	0.280	11	9	0.054	0.200	0.200	7	5	0.029	0.234	0.235	13	12	0.051
1	0.234	0.335	15	14	0.073	0.182	0.182	7	6	0.028	0.233	0.233	13	12	0.050
2	0.215	0.30	14	13	0.062	0.195	0.412	5	4	0.064	0.230	0.230	15	13	0.046
3	0.214	0.312	16	16	0.067	0.178	0.192	10	10	0.0342	0.227	0.227	13	12	0.048
4	0.212	0.317	18	16	0.060	0.177	0.182	8	7	0.028	0.215	0.215	13	12	0.043
5	0.212	0.335	19	18	0.067	0.175	0.189	9	8	0.0294	0.215	0.215	13	12	0.043
6	0.210	0.333	18	16	0.062	0.185	0.201	7	6	0.032	0.214	0.214	13	12	0.042
Mean	0.219	0.317	15.9	14.4	**0.067**	0.185	0.226	7.6	6.6	**0.035**	0.223	0.223	13.4	12.1	**0.053**
Dev.	0.010	0.018	2.8	3.0	0.005	0.009	0.077	1.6	1.8	0.012	0.009	0.009	3.0	0.4	0.004

poorer results compared to those obtained with the training data, which is attributale to the suboptimal predictions produced by the model.

Concerning the electric demand dataset, in Tables X, the results indicate significant differences among the XAI techniques evaluated. SHAP consistently achieves the highest global metrics, with an average RExQUAL value of 0.052, reflecting superior explanation quality. RULEx shows intermediate performance, identifying fewer key features (average of 20.8) and resulting in a RExQUAL of 0.041. In contrast, LIME exhibits the lowest performance, with a substantially lower average RExQUAL value of 0.033, indicating less effective explanations. On the other hand, for the SAGRA dataset, it is worth noting that, for the training data, the RULEx XAI technique demonstrated substantially better performance than the other XAI methods. However, in this case, the SHAP technique yields slightly superior average global metrics and a RExQUAL value of 0.067. In contrast, RULEx shows lower performance, with average support and confidence values of 0.185 and 0.226, respectively, yielding a mean RExQUAL of 0.035. Notably, LIME displays the least favorable outcomes, particularly evident in its significantly lower RExQUAL score of 0.045, despite a slightly higher mean support of 0.225. These findings indicate that while SHAP provides more reliable and interpretable explanations, RULEx and LIME struggle to achieve comparable levels of explanation quality, underscoring the variability in effectiveness among different XAI techniques in the context of the SAGRA dataset.

VI. CONCLUSIONS AND FUTURE WORK

This paper has introduced RExQUAL, a measure designed to evaluate and compare the quality of explanations provided by XAI methods. RExQUAL provides a quantitative basis for assessing the interpretability and reliability of existing XAI techniques. By establishing a standardized approach for comparison, RExQUAL facilitates the identification of the most effective methods for specific applications, thus advancing the overall field of XAI. The experiments have been conducted comparing three XAI techniques: SHAP, RULEx, and LIME. The results indicate that the RExQUAL metric effectively evaluates these XAI methods, significantly outperforming a random baseline. Notably, as the forecasting horizon extends further into the future, the quality of explanations diminishes due to increased prediction errors reflected in the generated ARs. Specifically, RULEx demonstrates markedly superior performance with multivariate time series data compared to the SHAP technique, while their performances are comparable in univariate datasets. In contrast, LIME yields the least favorable results among the evaluated methods. However, the proposed methodology has several

limitations that must be acknowledged. Firstly, the effectiveness of RExQUAL heavily relies on the specific rule-based method employed, potentially restricting its applicability across various scenarios. Secondly, calculating global measures incurs significant computational costs, which may hinder analysis efficiency. Moreover, the current rule-based approach does not support multi-step forecasting, limiting its ability to capture complex temporal dependencies across multiple time frames simultaneously. Last, RExQUAL lacks explicit statistical measures for direct comparison with other methods, as such measures have not been established. This limitation hinders the ability to assess the relative effectiveness of RExQUAL.

While the current experimentation focuses on time series data, future investigations could explore the adaptability of the proposed method to other data types, such as categorical data or images. Furthermore, RExQUAL is based on the extraction of ARs, and beyond the Apriori algorithm used in the experimental section, the landscape of AR algorithms is extensive, including variations such as FP-Growth and Eclat, each with its strengths and limitations. Future work aims to expand the comparative analysis to cover other methods such as BayLIME, SMILE or S-LIME or comparing to image attribution methods such as those in [7], [8], [30]. Moreover, image-based attribution techniques [28], [30], [39] could be applied and compared using RExQUAL Additionally, future work could include different AR algorithms to evaluate their performance within the proposed XAI framework. Finally, the proposed approach could be adapted to efficiently handle large-scale datasets by leveraging distributed computing techniques or GPU acceleration in Big Data contexts. Lastly, the global evaluation of ARs is crucial for the practical applicability of the proposed methodology. Research could focus on optimizing the computational efficiency of metric calculations, ensuring that the methodology remains scalable and applicable to large-scale datasets.

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